

$$1) \sqrt[3]{-27} + \sqrt[5]{-1} = \sqrt[3]{(-3)^3} + (-1) = -3 - 1 = -4$$

$$2) \frac{3^{-1} \cdot 4^{0.5}}{3^{-2} \cdot 2^{-1}} = \frac{3^{-1}}{3^{-2}} \cdot \frac{4^{0.5}}{2^{-1}} = 3^{-1-(-2)} \cdot \frac{\sqrt{4}}{2^{-1}} =$$

$$= 3 \cdot \frac{2}{\frac{1}{2}} = 3 \cdot 2 \cdot \frac{2}{1} = 12.$$

$$3) \log_8 4 + \log_8 32 - \log_8 2 =$$

$$= \log_8 (4 \cdot 32) - \log_8 2 = \log_8 \left(\frac{4 \cdot 32}{2} \right) =$$

$$= \log_8 64 = \log_8 (8^2) = 2.$$

$$\frac{a^n}{a^m} = a^{n-m}$$

$$\sqrt[n]{a} = a^{\frac{1}{n}}$$

$$a^{-1} = \frac{1}{a}$$

$$a^{-n} = \frac{1}{a^n}$$

$$\log_a x + \log_a y = \log_a (x \cdot y)$$

$$\log_a x - \log_a y = \log_a \left(\frac{x}{y} \right)$$

$$\log_a (x^n) = n \cdot \log_a x$$

$$x, y > 0$$

$$\log_a(a^n) = n$$

$$4) (\pi+1)^2 - (\pi-1)^2 =$$

$$= [(\pi+1) - (\pi-1)] \cdot [(\pi+1) + (\pi-1)] =$$

$$= (\cancel{\pi}+1-\cancel{\pi}+1) \cdot (\pi+\cancel{1}+\pi-\cancel{1}) = 4\pi$$

5) Niech n - nieparzysta l. całkowita

Nyknąć, że $12 \mid (3n^2 + 6n + 15)$, czyli, że

$$\underline{3n^2 + 6n + 15 = 12p}, \quad p \in \mathbb{Z}$$

David: n - nieparzyste, więc dla pewnego $k \in \mathbb{Z}$ $n = 2k+1$

Wtedy $L = 3n^2 + 6n + 15 = 3 \cdot (2k+1)^2 + 6(2k+1) + 15 =$

$$= 3 \cdot (4k^2 + 4k + 1) + 12k + 6 + 15 = 12k^2 + 12k + 3 + 12k + 21 =$$

$$= \underline{12k^2 + 24k + 24} = 12 \cdot \underbrace{(k^2 + 2k + 2)}_{=p} = 12p, \quad p = k^2 + 2k + 2 \in \mathbb{Z}$$

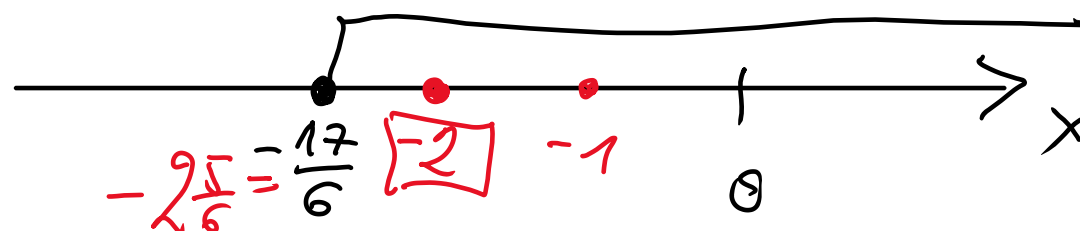
6) $\frac{7+2x}{5} \geq \frac{4}{15} / \cdot 15$ Znajdź najmniejszą l. całkow. spełniającą tę nierówność

$$3(7+2x) \geq 4$$

$$21+6x \geq 4$$

$$6x \geq -17$$

$$x \geq -\frac{17}{6}$$



7) $(x^2-4)(x^2-9)=0 \Leftrightarrow x^2=4 \vee x^2=9$
 $x=2 \vee x=-2 \quad x=3 \vee x=-3$

$$2 + (-2) + 3 + (-3) = \underline{\underline{0.}}$$

8) Dla $x \in \mathbb{R} \setminus \{-2; 0; 2\}$

$$\frac{4}{x^2+2x} : \frac{5x}{4-x^2} = \frac{4}{x^2+2x} \cdot \frac{4-x^2}{5x} = \frac{4}{x(x+2)} \cdot \frac{(2-x)(2+x)}{5x} = \frac{8-4x}{5x^2}$$

9)

$$x(3x-2) < 1$$

$$\underline{3x^2 - 2x - 1 < 0}$$

$$\Delta = 4 - 4 \cdot 3 \cdot (-1) = 16$$

$$\sqrt{\Delta} = 4$$

$$x_1 = \frac{-b - \sqrt{\Delta}}{2a}$$

$$x_2 = \frac{-b + \sqrt{\Delta}}{2a}$$

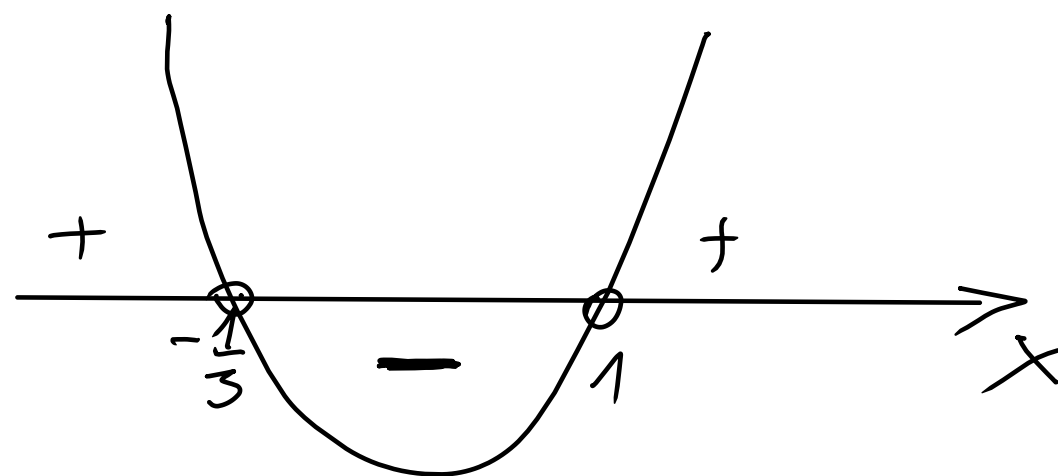
$$x_1 = \frac{2-4}{6} = -\frac{1}{3}$$

$$x_2 = \frac{2+4}{6} = 1$$

$$\underline{x \in \left(-\frac{1}{3}; 1\right)}$$

$$x^2 \leq 0$$

$$x = 0$$



10)

x - cena cegiełki

y - cena betonu

$$\begin{cases} 8x + 4y = 82 \quad | :2 \\ 3x + 9y = 60 \quad | :3 \end{cases}$$

$$\begin{cases} 4x + 2y = 41 \\ x + 3y = 20 \quad | \cdot (-4) \end{cases}$$

$$\begin{cases} 4x + 2y = 41 \\ -4x - 12y = -80 \end{cases}$$

$$= -10y = -39 \quad |: (-10)$$

$$y = 3,9$$

$$x = 20 - 3y$$

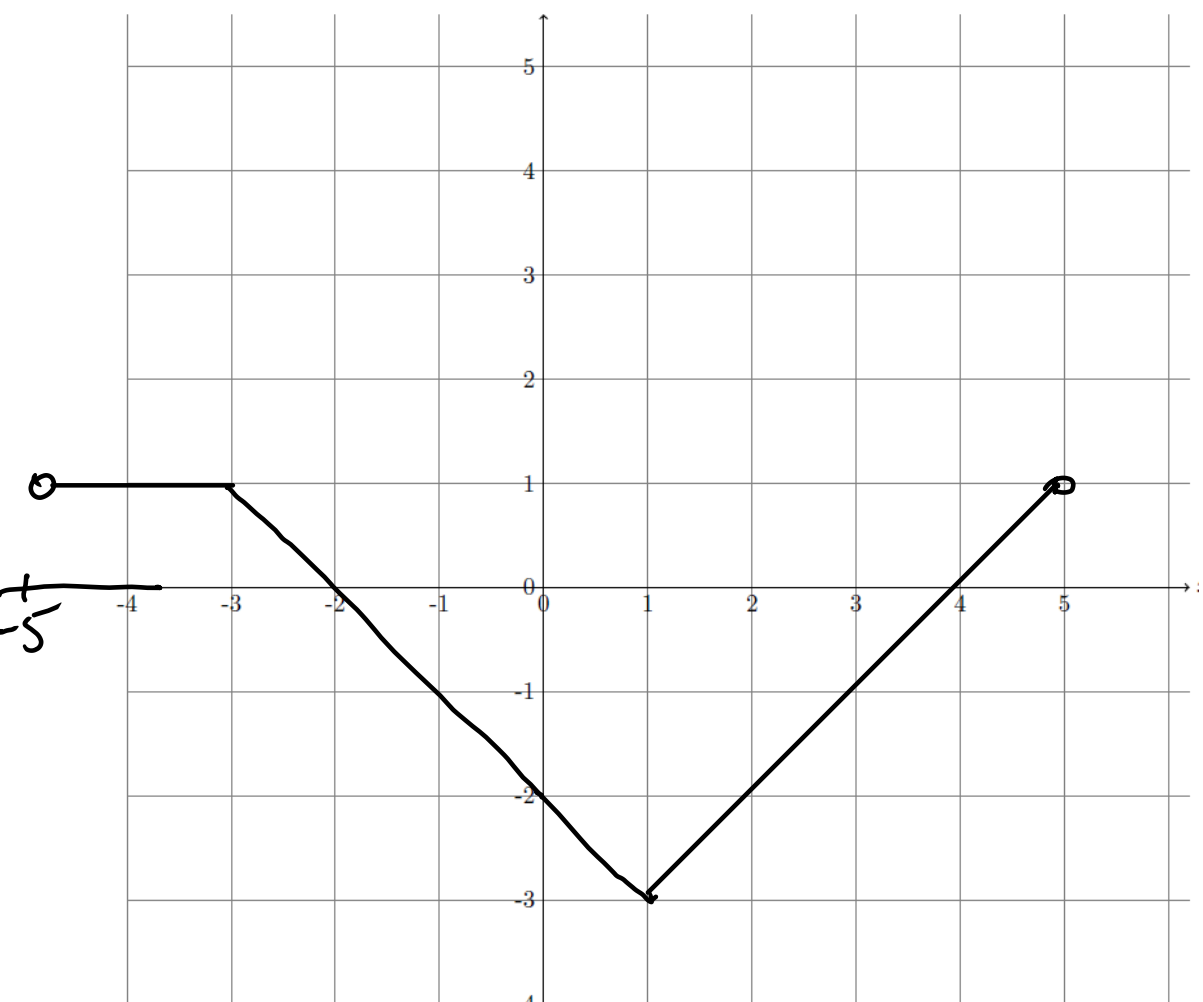
$$x = 20 - 11,7$$

$$\underline{x = 8,3}$$

$\langle a; b \rangle$

11)

$$f(x) = \begin{cases} 1 & \text{dla } x \in (-5; -3] \\ -x-2 & \text{dla } x \in (-3; 1] \\ x-4 & \text{dla } x \in (1, 5) \end{cases}$$



$$ZW = [-3; 1]$$

$$f(x) < 0 \text{ dla } x \in (-2; 4)$$

$$x \in (-5; -3]$$

$$12) \quad f(x) = 5x + m^2 + 6m, \quad m \in \mathbb{R}$$

Oblicz m wiedząc, że wykres przechodzi przez punkt $(0; -9)$

$$f(x) = y \quad \text{Dla } x = 0, y = -9$$

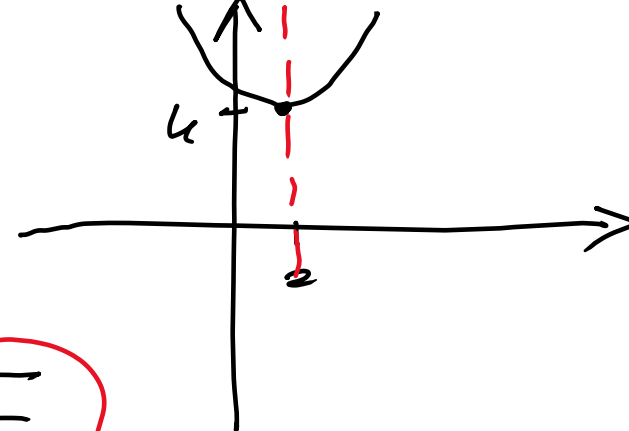
$$-9 = 5 \cdot 0 + m^2 + 6m$$

$$m^2 + 6m + 9 = 0$$

$$(m+3)^2 = 0$$

$$\underline{m = -3}$$

$$a=2, p=2, q=4$$



$$13.) f(x) = 2(x-2)^2 + 4$$

13.1) a) f. ma 2 miejsca zerowe

P

F

b) oś symetrii jest prosta $x=2$

P

F

13.2) postać ogólna to $f(x) = ax^2 + bx + c$

$$f(x) = 2 \cdot (x^2 - 4x + 4) + 4$$

$$f(x) = 2x^2 - 8x + 8 + 4$$

$$f(x) = 2x^2 - 8x + 12$$

$$14) a_n = 32 \cdot \left(\frac{1}{2}\right)^{n-5}, n \geq 1$$

$$14.1) a_3 = 32 \cdot \left(\frac{1}{2}\right)^{3-5} = 32 \cdot \left(\frac{1}{2}\right)^{-2} = 32 \cdot 2^2 = 128$$

14.2) a) c.a. $i \nearrow$
 b) c.a. $i \searrow$

c) c.g. $i \nearrow$
d) c.g. $i \searrow$

$$a_n = 32 \cdot \left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{2}\right)^{-5} = 32 \cdot 2^5 \cdot \left(\frac{1}{2}\right)^n = 32^2 \cdot \left(\frac{1}{2}\right)^n = \text{stato}_+ \cdot \left(\frac{1}{2}\right)^n$$

15) a_n - c.a. $n \geq 1$

$$S_{10} = 90, \quad \frac{a_1 + a_2 + a_3}{3} = 10$$

$$\begin{cases} \frac{2a_1 + 9r}{2} \cdot \cancel{10}^5 = 90 \\ \frac{a_1 + a_1 + 6r + a_1 + 7r}{3} = 10 \end{cases}$$

$$\begin{cases} (2a_1 + 9r) \cdot 5 = 90 \quad | :5 \\ \frac{3a_1 + 13r}{3} = 10 \quad | \cdot 3 \end{cases} \rightarrow$$

$$\begin{cases} 2a_1 + 9r = 18 \quad | \cdot (-3) \\ 3a_1 + 13r = 30 \quad | \cdot 2 \end{cases}$$

$$a_n = a_1 + (n-1) \cdot r$$

$$S_n = \frac{a_1 + a_n}{2} \cdot n$$

$$S_n = \frac{2a_1 + (n-1) \cdot r}{2} \cdot n$$

$$\begin{array}{l} \begin{cases} -6a_1 - 27r = -54 \\ + 2 \quad 6a_1 + 26r = 60 \end{cases} \\ \hline -r = 6 \\ \underline{r = -6} \end{array}$$

$$2a_1 - 54 = 18$$

$$2a_1 = 72$$

$$\underline{a_1 = 36}$$

!

$$\bar{X} = \frac{X_1 + X_2 + \dots + X_n}{n}$$

16) ciąg $(x, x+4, 2x+8)$ o wyrazach dodatnich jest geometryczny.

Wówczas tego ciągu jest równy

a) -4 , b) -1 , c) 2 , d) 4

! Liczby x, y, z tworzą c. geometryczny, gdy

$$\frac{y}{x} = \frac{z}{y}$$

U mes

$$\frac{x+4}{x} = \frac{2x+8}{x+4}$$

$$\frac{x+4}{x} = \frac{2(x+4)}{x+4}$$

$$\frac{x+4}{x} = 2 = q$$

17) $\alpha \in (0; 90^\circ)$ $\sin \alpha + \sin^2 \alpha = \cos \alpha - \cos^2 \alpha + 1$



For $\tan \alpha$ part ndms

a) $\frac{1}{2}$

b) $\frac{\sqrt{2}}{2}$

c) 1

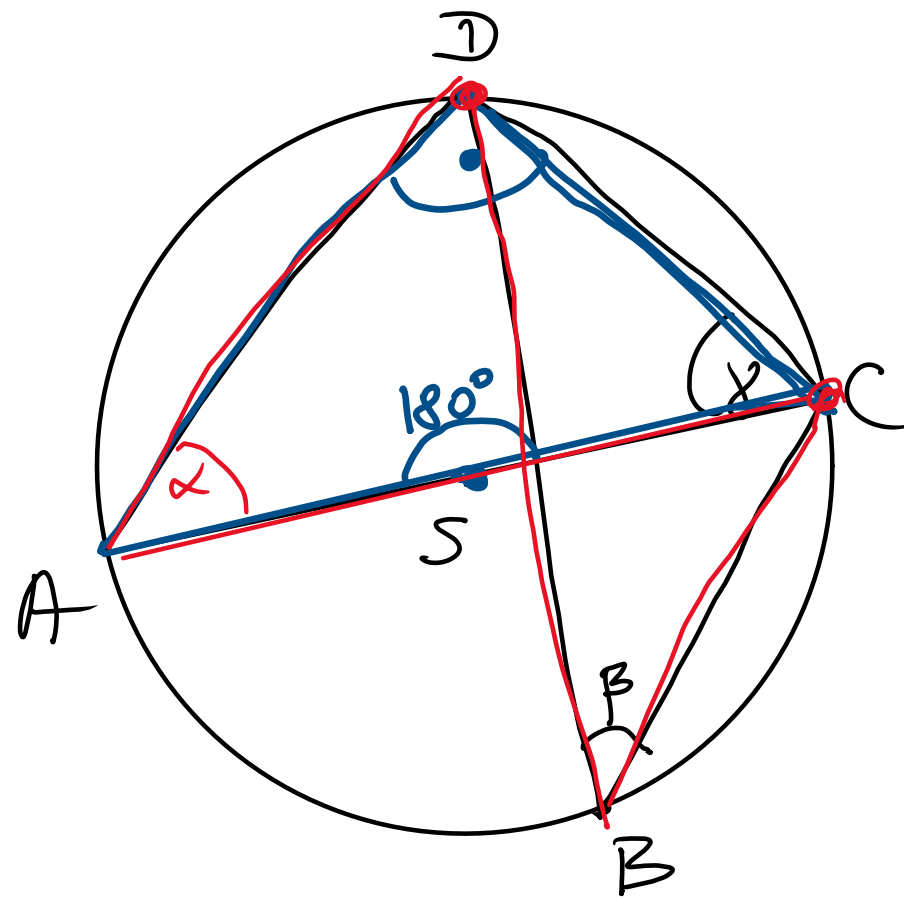
d) 2

$$\sin \alpha + \underbrace{\sin^2 \alpha + \cos^2 \alpha}_1 = \cos \alpha + 1$$

$$\sin \alpha = \cos \alpha \quad | : \cos \alpha$$

$$\frac{\sin \alpha}{\cos \alpha} = 1$$

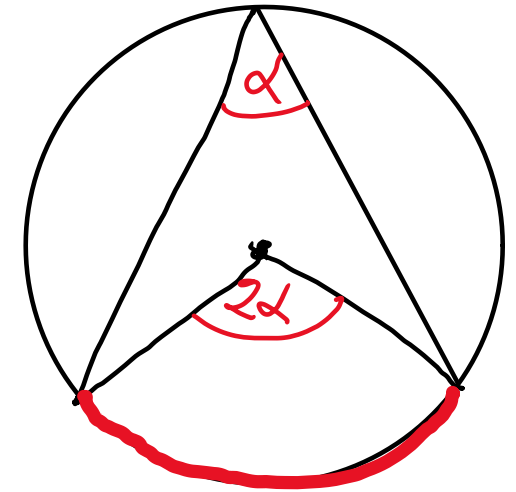
$$\underline{\tan \alpha = 1}$$



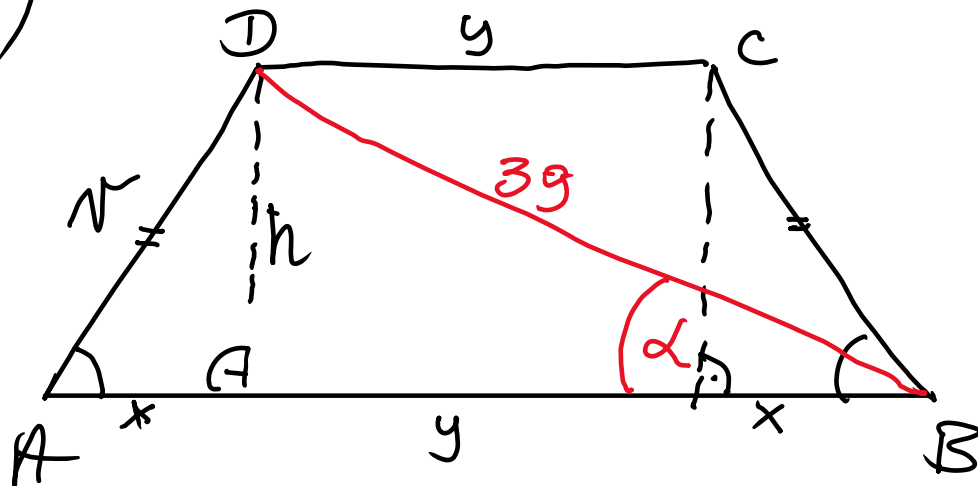
Suma $\beta + \gamma$ jest równa

- a) 60°
- b) 90°
- c) 120°
- d) 135°

$$\alpha = \beta$$

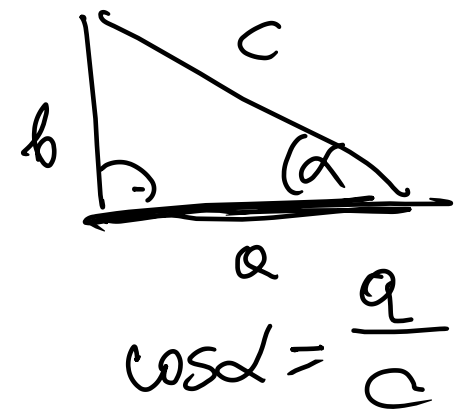


19)



$$\cos \alpha = \frac{12}{13}$$

$$|AB| - |CD| = 40$$



$$\cos \alpha = \frac{a}{c}$$

Oblicz $|AD| = ?$

$$|AB| = 2x + y$$

$$|CD| = y$$

$$2x + y - y = 40$$

$$\underline{x = 20}$$

$$\cos \alpha = \frac{x+y}{39} = \frac{12}{13}$$

$$\frac{20+y}{39} = \frac{12}{13} \quad | \cdot 39$$

$$20+y = 36$$

$$\underline{y = 16}$$

$$|AB| = 2 \cdot 20 + 16 = 56$$

2. dw. cosinusowa

$$\cancel{r^2} = r^2 = |AB|^2 + |BD|^2 - 2 \cdot |AB| \cdot |BD| \cdot \cos \alpha$$

$$r^2 = \dots$$

